

Fiber Optics - PMD report

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I. INTRODUCTION

Ideal single-mode fibers are known to support two orthogonal modes, $LP_{0,1}^{(x)}$ and $LP_{0,1}^{(y)}$, which, thanks to the circular symmetry of an optical fiber section, are degenerate. In other words, the propagation constant inside the fiber does not depend on the direction of the polarization.

However, real fibers are not perfectly symmetric, and this causes a difference in the propagation constants of the two orthogonal modes, making them non-degenerate: this phenomenon is called polarization mode dispersion (PMD). Several phenomena can induce asymmetry in the fiber sections, and most of them are random and vary along the fiber, hence they cannot be accounted for *a priori*.

PMD can be described by the so-called differential group delay (DGD) $\Delta\tau$, namely the time delay between the arrival of two orthogonally-polarized pulses at the end of the fiber.

In general, the state of polarization (SOP) at a certain position along the fiber depends on the frequency. PMD can be thus described both space- and frequency-wise. In the latter case, a remarkable mathematical tool used to describe this type of dispersion is the PMD vector $\bar{\Omega}(\omega)$, whose absolute value represents the DGD, and whose direction indicates the principle states of polarizations (PSP), namely particular SOPs that do not vary with frequency.

The PMD vector is related to the unit Stokes vector $\hat{s}$ (representing the SOP) as

$$\frac{d\hat{s}}{d\omega} = \bar{\Omega}(\omega) \times \hat{s}(\omega). \quad (1)$$

In general, because of the random nature of this type of dispersion, the PMD vector itself depends on frequency. Nonetheless, in the so-called polarization-maintaining fibers (PMF), whose symmetry is purposely altered in a controlled way, $\bar{\Omega}(\omega) = \text{const}$. Because of (1), the velocity of the unit Stokes vector is always orthogonal to both the PMD vector and $\hat{s}$ itself, therefore as it varies with frequency, it rotates about $\bar{\Omega}$, tracing a circular trajectory.

In order to estimate the DGD, the function $T(\omega)$ is introduced: given an arbitrary $\hat{p}$ unit vector, the function $T(\omega)$ is

$$T(\omega) = \frac{1}{2}(1 + \hat{s}(\omega) \cdot \hat{p}), \quad (2)$$

where $\hat{p}$ is an arbitrary unit vector. If the fiber is polarization-maintaining, then $T(\omega)$ is periodic with period $2\pi/\Delta\tau$.

Conversely, in a standard fiber $T(\omega)$ is a random process. A viable approach to measure the DGD is to use level crossing-rate analysis. By measuring at various levels ν_i the number of times $n(\nu_i)$ that $T(\omega)$ crosses the level ν_i per unit frequency band, the average DGD is estimated as

$$\langle \Delta\tau \rangle = \frac{2 \sum_i n(\nu_i) \sqrt{\nu_i(1-\nu_i)}}{\sum_i \nu_i(1-\nu_i)}. \quad (3)$$

The experimental setup is clarified in the following section. Next, in Section III the obtained results are presented and commented on. Finally, in Section IV some conclusions are drawn from our analysis and overall laboratory experience.

II. EXPERIMENTAL SETUP

This laboratory experience was divided into two parts. In the first part, shown in Figure 1, an external-cavity laser (ECL) with a tunable wavelength is used to generate monochromatic light. The emitted beam is then polarized by a polarization state generator (PSG), which is part of a polarimeter. Next, the signal travels along a very short PMF, which is about half a meter long and is inserted inside a CD case.

Finally, light reaches a polarization state analyzer (PSA), which measures the polarization, calculates the unit Stokes vector and plots it on a Poincaré sphere on its display. Just like the PSG, polarization state analysis is provided by the polarimeter. With this setup, one is able to vary the optical frequency of EM radiation and observe the effect on the final polarization state measured by the device.

Our task for the first part was to collect a sufficient amount of measurements at various optical frequencies to show that the fiber is polarization-maintaining, namely to verify that the unit Stokes vector that measures the SOP is periodic and rotates about a certain vector, which, as given by (1), is the PMD vector itself. Moreover, the DGD value was quantified by applying (2) to PMFs.

The setup of the second part of the experiment is quite similar to the first one, with the exception of the optical fiber. In this case, the fiber connected to the polarimeter was not polarization-maintaining. Instead, it was part of a ribbon cable, namely a group of optical fibers arranged onto a strip and covered together by the same plastic coating. This type of configuration helps increasing the fiber density, which is very important for easing deployment, and decreasing both cost and the required space. However, the coating applies a slight asymmetrical pressure on the fibers, causing some PMD. Nonetheless, the overall effect of this, as will be shown in the analysis, is marginal.

The second fiber, which is more sensitive to external fluctuations, is inserted into a box filled with foam, to reduce variations caused by temperature. Similarly to the first part, several measurements of the unit Stokes vector are obtained. As will be explained in the analysis section, because of the

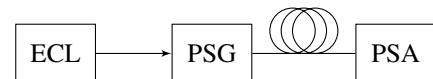


Fig. 1: Experimental setup

random nature of this type of dispersion in normal fibers, more data is required to have consistent results.

III. RESULTS

A. Polarization-maintaining fiber

The results obtained for the first fiber confirm the fact that it is actually polarization maintaining: in Fig. 2 the rotation of the Stokes vector along a circle in the Poincaré sphere is clearly visible.

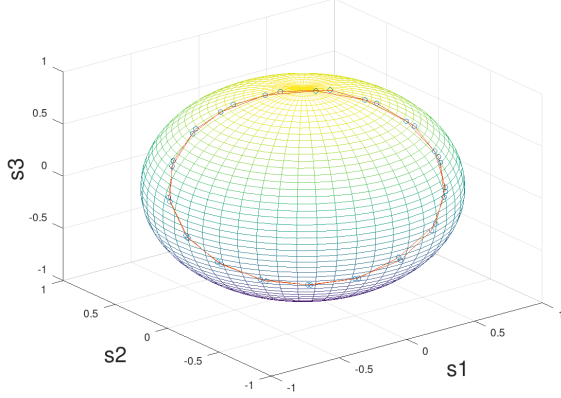


Fig. 2: Stokes parameter measurement for PMF sample.

As regards the DGD estimation, in this first case, as anticipated, it is possible to simply evaluate the period of $\hat{s}(\omega)$ ¹. We expect to find the same periodicity for all the 3 channels of measurement of $\hat{s}$, so, in order to average the measure over these channels, a randomized computation is preferred over the simple computation over the three components: the SOP is projected along a set of polarization vectors $\hat{p}$ chosen uniformly on the Poincaré sphere. The test consists in evaluating the period of the multiple projected functions, computed as $(\hat{s}(\omega) - \hat{p}) - \mu$, where μ is the sample mean of $\hat{s}(\omega) - \hat{p}$, so that the function, that will be sinusoidal in ω , will have, over the various $\hat{p}$, 0 mean. With the choice of 100 $\hat{p}$ vectors, by averaging the angular frequency period Θ over the whole set of projections, we estimate the DGD for the PMF sample as

$$\langle \Delta\tau \rangle_{PM} = \frac{2\pi}{\Theta} = 2.30 \text{ ps.}$$

Moreover, since the DGD vector $\bar{\Omega}$ is constant in this case, we can use the law described in (1) to develop a least-squares estimate of $\bar{\Omega}$. By finding its estimate, the PSP are calculated as the direction of the vector and its opposite, and the DGD can be computed as its norm. The least-square estimation harnesses the fact that the law of evolution is linear in $\bar{\Omega}$, and the derivatives of the components of $\hat{s}$ are readily computed as incremental ratios. If we consider a gaussian IID noise on the measurements, the least squares (LS) vector reads:

$$\bar{\Omega}^{LS} = \begin{pmatrix} -1.086 \\ -1.567 \\ +1.056 \end{pmatrix} \text{ ps.}$$

¹Note that the period is not intended in the usual sense: it obviously measures a periodicity in *angular frequency*.

Its norm, which denotes the DGD, is not exactly equal to the one computed with the period estimation, probably because of the small set of measures.

$$\langle \Delta\tau \rangle_{PM}^{LS} = |\bar{\Omega}^{LS}| = 2.18 \text{ ps.}$$

B. Standard single-mode fiber

In the case of the second fiber sample, the polarization is no more following a circle on the Poincaré sphere, as shown in Fig. 3.

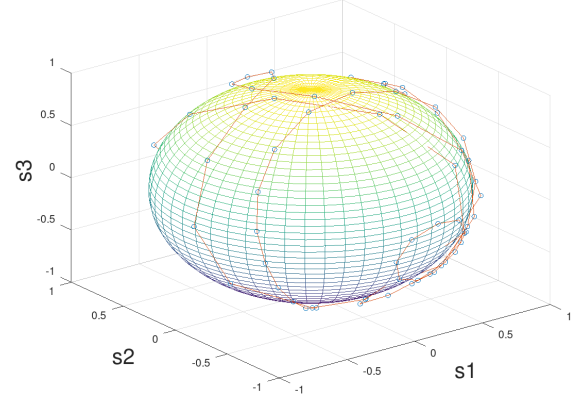


Fig. 3: Stokes parameter measurement for the standard fiber sample.

In this situation the $T(\omega)$ function is not sinusoidal, and in general a simple estimation of the DGD is not available. However, using the method described in (3), we select a number of levels equal to 5, and conduct a randomized analysis over 100 different $\hat{p}$ vectors. As an example, in Fig. 4 two different projections for T are plotted against the frequency.

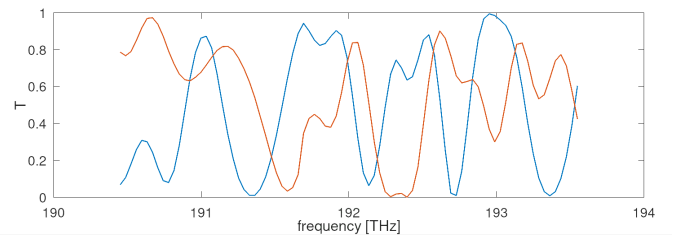


Fig. 4: T function for two different $\hat{p}$.

The value for the DGD is then estimated as

$$\langle \Delta\tau \rangle_{STD} = 1.97 \text{ ps.}$$

A technique used in the analysis part is the generation of uniformly distributed random vectors in the Poincaré sphere. In order to have an uniform distribution over the spherical surface it is not sufficient to assign three cartesian uniform distributed coordinates: in fact this procedure gives a set of vector uniformly distributed in a cube, and the subsequent normalization gives a spherical distribution with lower density in correspondence to the cube facets. Even further, the problem is not adequately addressed by using a uniform distribution

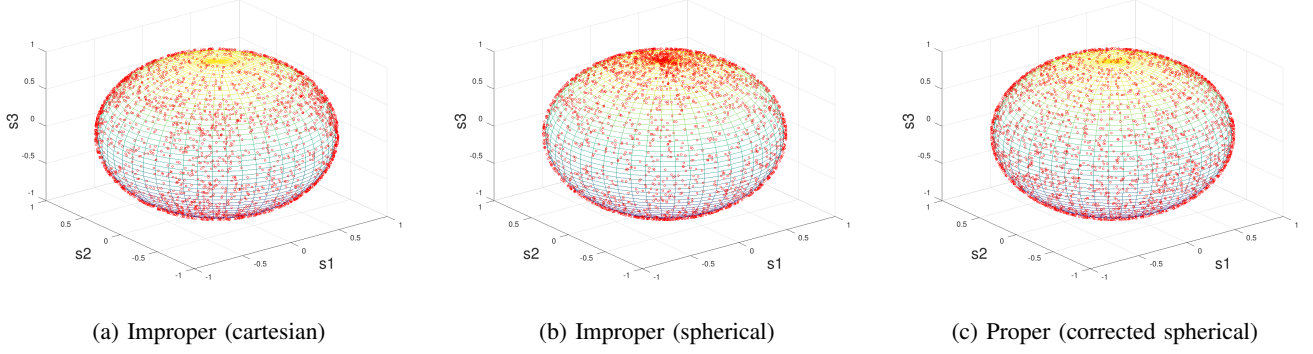


Fig. 5: Effect of improper parametrization of $\hat{p}$ vectors.

of the angles of the spherical coordinates parametrization. A detailed discussion is beyond the scope of this report, in fact, a proper distribution is given by computing the usual spherical coordinates angles θ, ϕ from two uniformly distributed variables ξ_1, ξ_2 by means of the following transformation

$$\begin{aligned}\theta &= \arccos(1 - 2\xi_1) \\ \phi &= \xi_2\end{aligned}$$

The above result is deduced using the theorem of universality of uniform distribution.

In Fig. 5 the three methods are used to assign 4000 random vectors in the Poincaré sphere. As can be seen, only Fig. 5(c) shows a uniform distribution.

In this experiment, having a uniform distribution is not actually critical: an improper parametrization induces, over 100 random vectors, a DGD deviation which is negligible in the case of PMF, and in the order of 0.1 ps in the standard fiber case. However, there are cases of Monte Carlo analysis in which the accuracy in the uniform distribution is paramount.

IV. CONCLUSIONS

The analysis of the data collected from the polarization state analyzer was performed by taking two different approaches, which were described in the introduction and expanded on in the results section.

The polarization-maintaining fiber, whose length was of few decimeters, was estimated to have about 2.30 ps of DGD. Considering the length of this fiber, the result is coherent with the large values typical of PM fibers, namely few picoseconds per meter. Furthermore, the spatial distribution of our data clearly shows that by varying the frequency, the unit Stokes vector rotates about the PSP, as expected from a PMF (given that $\hat{\Omega}(\omega) = \text{const}$).

As regards the single-mode fiber, it is trickier to compare the obtained DGD ($\Delta\tau = 1.97$ ps) to the typical values in the market, which can be as low as 0.01 ps/ $\sqrt{\text{km}}$ for high-quality fibers, but is more commonly around 0.1 ps/ $\sqrt{\text{km}}$. Given the result, which is relatively large for a single-mode fiber, we suspect that the fiber is characterized by a larger DGD coefficient. Since the length of the fiber used in the lab is about 12 km, it corresponds to 0.569 ps/ $\sqrt{\text{km}}$, which is not

far from some of the common values that appear in the ITU-T G.652 standard (0.5 ps/ $\sqrt{\text{km}}$).

Moreover, the large value may also partly be caused by the flat ribbon used to group multiple fibers. Even though the contribution of this should be small, it may explain why, supposing that the standard performance of the fiber is similar to those previously mentioned and found in the ITU-T standard, the value is still a little larger.

What is more probable, though, is that the data that we have collected in the lab are not sufficient to obtain consistent results of a random phenomenon such as polarization-mode dispersion in a regular fiber. While for the polarization-maintaining fiber the small number of measures was enough because of the periodicity of $T(\omega)$, in the second case the variations are unpredictable, and the obtained DGD may be overestimated.